

DestinE ClimateDT Parameters

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Parameters forming part of the Climate Change Adaptation Digital Twin (DT) are detailed here.

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Data availability
See the [Climate DT Phase 1 data catalogue](#) page for more details on data availability and accessing the data.

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Additional keys
Please note that additional keys are often necessary in addition to the base parameter definition below. For example for:

- statistical processing (e.g. time interval of processing)
- level information (e.g. a given pressure level, say 300 hPa)

Legend:

Parameter ready for GRIB2 archiving

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1. Instantaneous encodings

1.1. Fields on a single level or surface

1.1.1. Atmosphere (levtype sfc)

Time resolution: Hourly (time=0000/0100/...)

param Id	name	short Name	units	discipline	parameter Category	parameter Number	type Of Level	level	type Of Statistical Processing	local Tables Version	comments
78	Total column cloud liquid water	tclw	kg m**-2	0	1	69	entireAtmosphere	-			model=ICON/IFS-NEMO/IFS-FESOM HEALPIX
79	Total column cloud ice water	tciw	kg m**-2	0	1	70	entireAtmosphere	-			model=ICON/IFS-NEMO/IFS-FESOM HEALPIX
130	Temperature	t	K	0	0	0	surface	-			model=ICON HEALPIX
134	Surface pressure	sp	Pa	0	3	0	surface	-			model=ICON/IFS-NEMO/IFS-FESOM HEALPIX
137	Total column vertically-integrated water vapour	tcwv	kg m**-2	0	1	64	entireAtmosphere	-			model=ICON/IFS-NEMO/IFS-FESOM HEALPIX
141	Snow depth	sd	m of water equivalent	0	1	254	surface	-		1	model=IFS-NEMO/IFS-FESOM HEALPIX. Depth of snow with conversion to water. ECMWF unit.
148	Charnock	chnk	Numeric	10	0	76	surface	-			model=IFS-NEMO/IFS-FESOM HEALPIX

151	Mean sea level pressure	msl	Pa	0	3	0	meanSea	-			model=ICON/IFS-NEMO/IFS-FESOM HEALPIX
159	Boundary layer height	blh	m	0	3	18	surface	-			model=IFS-NEMO/IFS-FESOM HEALPIX
164	Total cloud cover	tcc	(0 - 1)	0	6	192	entireAtmosphere	-		1	model=IFS-NEMO/IFS-FESOM HEALPIX. ECMWF unit.
165	10 metre U wind component	10u	m s ⁻¹	0	2	2	heightAboveGround	10			model=ICON/IFS-NEMO/IFS-FESOM HEALPIX
166	10 metre V wind component	10v	m s ⁻¹	0	2	3	heightAboveGround	10			model=ICON/IFS-NEMO/IFS-FESOM HEALPIX
167	2 metre temperature	2t	K	0	0	0	heightAboveGround	2			model=ICON/IFS-NEMO/IFS-FESOM HEALPIX
168	2 metre dewpoint temperature	2d	K	0	0	6	heightAboveGround	2			model=ICON/IFS-NEMO/IFS-FESOM HEALPIX
186	Low cloud cover	lcc	(0 - 1)	0	6	193	lowCloudLayer	-		1	model=IFS-NEMO/IFS-FESOM HEALPIX. ECMWF unit.
187	Medium cloud cover	mcc	(0 - 1)	0	6	194	mediumCloudLayer	-		1	model=IFS-NEMO/IFS-FESOM HEALPIX. ECMWF unit.
188	High cloud cover	hcc	(0 - 1)	0	6	195	highCloudLayer	-		1	model=IFS-NEMO/IFS-FESOM HEALPIX. ECMWF unit.
235	Skin temperature	skt	K	0	0	17	surface	-			model=ICON/IFS-NEMO/IFS-FESOM HEALPIX
228164	Total Cloud Cover	tcc	%	0	6	1	entireAtmosphere	-			model=ICON HEALPIX. WMO unit.
260048	Total precipitation rate	tptrate	kg m ⁻² s ⁻¹	0	1	52	surface	-			model=ICON/IFS-NEMO/IFS-FESOM HEALPIX

Time resolution: Daily (date=YYYYMMDD,time=0000)

param Id	name	short Name	units	discipline	parameter Category	parameter Number	type Of Level	level	type Of Statistical Processing	local Tables Version	comments
129	Geopotential	z	m ² s ⁻²	0	3	4	surface	-			model=IFS-NEMO/IFS-FESOM HEALPIX. Geopotential at surface gives orography. This output of constant fields is needed by applications.
172	Land-sea mask	lsm	(0 - 1)	2	0	0	surface	-			model=IFS-NEMO/IFS-FESOM HEALPIX. This output of constant fields is needed by applications.

1.2. Fields on pressure levels (levtype p l)

Time resolution: Hourly (time=0000/0100/...)

19 pressure levels: (1000, 925, 850, 700, 600, 500, 400, 300, 250, 200, 150, 100, 70, 50, 30, 20, 10, 5, 1)

param Id	name	short Name	units	discipline	parameter Category	parameter Number	type Of Level	level	type Of Statistical Processing	local Tables Version	comments
60	Potential vorticity	pv	K m**2 kg**-1 s**-1	0	2	14	isobaricInhPa	<as-above>			model=IFS-NEMO/IFS-FESOM HEALPIX
129	Geopotential	z	m**2 s**-2	0	3	4	isobaricInhPa	<as-above>			model=ICON/IFS-NEMO/IFS-FESOM HEALPIX
130	Temperature	t	K	0	0	0	isobaricInhPa	<as-above>			model=ICON/IFS-NEMO/IFS-FESOM HEALPIX
131	U component of wind	u	m s**-1	0	2	2	isobaricInhPa	<as-above>			model=ICON/IFS-NEMO/IFS-FESOM HEALPIX
132	V component of wind	v	m s**-1	0	2	3	isobaricInhPa	<as-above>			model=ICON/IFS-NEMO/IFS-FESOM HEALPIX
133	Specific humidity	q	kg kg**-1	0	1	0	isobaricInhPa	<as-above>			model=ICON/IFS-NEMO/IFS-FESOM HEALPIX
135	Vertical velocity	w	Pa s**-1	0	2	8	isobaricInhPa	<as-above>			model=ICON/IFS-NEMO/IFS-FESOM HEALPIX This is in "Pa s-1" not "m s-1".
157	Relative humidity	r	%	0	1	1	isobaricInhPa	<as-above>			model=ICON/IFS-NEMO/IFS-FESOM HEALPIX
246	Specific cloud liquid water content	clwc	kg kg**-1	0	1	83	isobaricInhPa	<as-above>			model=ICON/IFS-NEMO/IFS-FESOM HEALPIX

1.3. Fields on model levels

1.3.1. Sea-ice/snow/soil (levtype soI)

Time resolution: Hourly (time=0000/0100/...)

5 model levels: (1, ..., 5)

param Id	name	short Name	units	discipline	parameter Category	parameter Number	type Of Level	level	type Of Statistical Processing	local Tables Version	comments
228141	Snow depth water equivalent	sd	kg m**-2	0	1	60	snow	<as-above>			model=IFS-NEMO/IFS-FESOM HEALPIX. Depth of snow with conversion to water. WMO unit.

1.4. Fields on height levels (levtype hI)

Time resolution: Hourly (time=0000/0100/...)

Currently 100m only. **These are only for height levels >10m to preserve backwards compatibility with longstanding model fields (2t, 10u, 10v,...)**

param Id	name	short Name	units	discipline	parameter Category	parameter Number	type Of Level	level	type Of Statistical Processing	local Tables Version	comments
228246	100 metre U wind component	100u	m s**-1	0	2	2	heightAboveGround	100			model=IFS-NEMO/IFS-FESOM HEALPIX
228247	100 metre V wind component	100v	m s**-1	0	2	3	heightAboveGround	100			model=IFS-NEMO/IFS-FESOM HEALPIX

2. Mean encodings

2.1. Fields on a single level or surface

2.1.1. Sea ice (levtype o2d)

Time resolution: 1 x daily (date=YYYYMMDD,time=0000)

param Id	name	short Name	units	discipline	parameter Category	parameter Number	type Of Level	level	type Of Statistical Processing	local Tables Version	comments
263000	Time-mean sea ice thickness	avg_sithick	m	10	2	1	iceLayerOnWater	-	0		model=ICC NEMO/IFS FESOM O2D
263001	Time-mean sea ice area fraction	avg_siconc	Fraction	10	2	0	iceLayerOnWater	-	0		model=ICC NEMO/IFS FESOM O2D
263002	Time-mean snow thickness over sea ice	avg_sisnthick	m	10	2	192	snowLayerOverIceOnWater	-	0	1	model=IFS FESOM O2D
263003	Time-mean eastward sea ice velocity	avg_siue	m s ⁻¹	10	2	4	iceLayerOnWater	-	0		model=ICC NEMO/IFS FESOM O2D
263004	Time-mean northward sea ice velocity	avg_sivn	m s ⁻¹	10	2	5	iceLayerOnWater	-	0		model=ICC NEMO/IFS FESOM O2D
263008	Time-mean sea ice volume per unit area	avg_sivol	m ³ m ⁻²	10	2	15	iceLayerOnWater	-	0		model=IFS NEMO/IFS FESOM O2D
263009	Time-mean snow volume over sea ice per unit area	avg_snvol	m ³ m ⁻²	10	2	16	snowLayerOverIceOnWater	-	0		model=ICC NEMO/IFS FESOM O2D
263021	Time-mean X-component of sea ice velocity	avg_six	m s ⁻¹	10	2	24	iceLayerOnWater	-	0		model=IFS NEMO/IFS FESOM O2D
263022	Time-mean Y-component of sea ice velocity	avg_siy	m s ⁻¹	10	2	25	iceLayerOnWater	-	0		model=IFS NEMO/IFS FESOM O2D

2.1.2. Ocean (levtype o2d)

Time resolution: 1 x daily (date=YYYYMMDD,time=0000)

param Id	name	short Name	units	discipline	parameter Category	parameter Number	type Of Level	level	type Of Statistical Processing	local Tables Version	comments
263100	Time-mean sea surface practical salinity	avg_sos	g kg ⁻¹	10	3	3	oceanSurface	-	0		model=IFS NEMO/IFS FESOM O2D
263101	Time-mean sea surface temperature	avg_tos	K	10	3	0	oceanSurface	-	0		model=IFS NEMO/IFS FESOM O2D
263114	Time-mean ocean mixed layer depth defined by sigma theta 0.03 kg m-3	avg_mlotst030	m	10	4	14	mixedLayerDepthByDensity	0.03	0		model=ICC O2D
263121	Time-mean vertically-integrated heat content in the upper 300 m	avg_hc300m	J m ⁻²	10	4	22	depthBelowSeaLayer	300	0		model=IFS NEMO/IFS FESOM O2D

263122	Time-mean vertically-integrated heat content in the upper 700 m	avg_hc700m	J m ⁻²	10	4	22	depthBelowSeaLayer	700	0		model=IFS-NEMO/IFS-FESOM O2D
263124	Time-mean sea surface height	avg_zos	m	10	3	1	oceanSurface	-	0		model=ICON-NEMO/IFS-FESOM O2D

2.2. Fields on model levels

2.2.1. Ocean (levtype o3d)

Time resolution: 1 x daily (date=YYYYMMDD,time=0000)

ICON: 72 model levels: (1, ..., 72)

NEMO: 75 model levels: (1, ..., 75)

FESOM: 69 model levels: (1, ..., 69)

param Id	name	short Name	units	discipline	parameter Category	parameter Number	type Of Level	level	type Of Statistical Processing	local Tables Version	comments
263500	Time-mean sea water practical salinity	avg_so	g kg ⁻¹	10	4	21	oceanModelLayer	<as-above>	0		model=ICON/IFS-NEMO/IFS-FESOM O3D
263501	Time-mean sea water potential temperature	avg_thetao	K	10	4	18	oceanModelLayer	<as-above>	0		model=ICON/IFS-NEMO/IFS-FESOM O3D
263505	Time-mean northward sea water velocity	avg_von	m s ⁻¹	10	4	24	oceanModelLayer	<as-above>	0		model=ICON/IFS-NEMO/IFS-FESOM O3D
263506	Time-mean eastward sea water velocity	avg_uoe	m s ⁻¹	10	4	23	oceanModelLayer	<as-above>	0		model=ICON/IFS-NEMO/IFS-FESOM O3D
263507	Time-mean upward sea water velocity	avg_wo	m s ⁻¹	10	4	27	oceanModel	<as-above>	0		model=ICON/IFS-NEMO/IFS-FESOM O3D

3. Accumulated encodings

3.1. Fields on a single level or surface

3.1.1. Atmosphere (levtype sfc)

Time resolution: Hourly accumulation (time=0000/0100/...)

param Id	name	short Name	units	discipline	parameter Category	parameter Number	type Of Level	level	type Of Statistical Processing	local Tables Version	comments
8	Surface runoff	sro	m	2	0	202	surface	-	1	1	model=IFS-NEMO/IFS-FESOM HEALPIX ECMWF unit
9	Sub-surface runoff	ssro	m	2	0	204	surface	-	1	1	model=IFS-NEMO/IFS-FESOM HEALPIX ECMWF unit
144	Snowfall	sf	m of water equivalent	0	1	198	surface	-	1	1	model=IFS-NEMO/IFS-FESOM HEALPIX ECMWF unit
146	Surface sensible heat flux	sshf	J m ⁻²	0	0	11	surface	-	1		model=ICON/IFS-NEMO/IFS-FESOM HEALPIX
147	Surface latent heat flux	slhf	J m ⁻²	0	0	10	surface	-	1		model=ICON/IFS-NEMO/IFS-FESOM HEALPIX
169	Surface short-wave (solar)	ssrd	J m ⁻²	0	4	7	surface	-	1		model=ICON/IFS-NEMO/IFS-FESOM HEALPIX

	radiation downwards										
175	Surface long-wave (thermal) radiation downwards	strd	J m**-2	0	5	3	surface	-	1		model=ICON/IFS-NEMO/IFS-FESOM HEALPIX
176	Surface net short-wave (solar) radiation	ssr	J m**-2	0	4	9	surface	-	1		model=ICON/IFS-NEMO/IFS-FESOM HEALPIX
177	Surface net long-wave (thermal) radiation	str	J m**-2	0	5	5	surface	-	1		model=ICON/IFS-NEMO/IFS-FESOM HEALPIX
178	Top net short-wave (solar) radiation	tsr	J m**-2	0	4	9	nominalTop	-	1		model=ICON/IFS-NEMO/IFS-FESOM HEALPIX
179	Top net long-wave (thermal) radiation	ttr	J m**-2	0	5	5	nominalTop	-	1		model=ICON/IFS-NEMO/IFS-FESOM HEALPIX
180	Time-integrated eastward turbulent surface stress	ewss	N m**-2 s	0	2	62	surface	-	1		model=IFS-NEMO/IFS-FESOM HEALPIX
181	Time-integrated northward turbulent surface stress	nsss	N m**-2 s	0	2	63	surface	-	1		model=IFS-NEMO/IFS-FESOM HEALPIX
182	Evaporation	e	m of water equivalent	0	1	199	surface	-	1	1	model=IFS-NEMO/IFS-FESOM HEALPIX ECMWF unit
212	TOA incident short-wave (solar) radiation	tisr	J m**-2	0	4	7	nominalTop	-	1		model=IFS-NEMO/IFS-FESOM HEALPIX
228	Total precipitation	tp	m	0	1	193	surface	-	1	1	model=IFS-NEMO/IFS-FESOM HEALPIX ECMWF unit
260654	Time-integrated eastward turbulent surface stress due to surface roughness	etsssr	N m**-2 s	0	2	66	surface	-	1		model=ICON HEALPIX
260655	Time-integrated northward turbulent surface stress due to surface roughness	ntsssr	N m**-2 s	0	2	67	surface	-	1		model=ICON HEALPIX

4. Maximum encodings
- There are no maximum encodings for this dataset.
5. Minimum encodings
- There are no minimum encodings for this dataset.

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